

KAIMANA UTAROM, Indonesia

WMO: 977600

Lat: 3.6439S

Lon: 133.6969E

Elev: 10

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	73.0	73.7	69.0	106.7	77.4	70.0	110.4	77.5	18.6	79.1	16.4	80.5	1.8	90	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	11.5	91.4	80.5	90.3	80.1	89.4	79.9	81.6	88.9	81.0	88.1	80.6	87.6	7.0	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.3	152.0	85.6	79.0	150.3	85.3	78.4	147.7	84.8	45.3	89.5	44.7	88.8	44.2	87.9	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.1	12.8	11.3	DB	70.3	96.1	5.0	3.6	66.7	98.7	63.8	100.8	60.9	102.8	57.3	105.4	
			WB	68.3	83.9	4.5	1.7	65.1	85.2	62.4	86.2	59.9	87.1	56.7	88.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.4	83.0	82.8	82.7	82.2	81.7	80.4	79.2	78.9	79.7	81.3	82.2	83.1
	DBStd	2.17	1.96	2.01	1.88	1.44	1.42	1.59	1.41	1.47	1.46	1.28	1.49	1.78
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11472	1022	918	1015	967	982	911	906	896	892	970	967	1026
	CDD65	5997	557	498	550	517	517	461	441	431	442	505	517	561
	CDH74	60892	6221	5500	5985	5520	5504	4360	3619	3303	3880	5185	5574	6241
CDH80	17771	2254	1986	2090	1837	1678	875	404	316	653	1457	1927	2294	
Wind	WSAvg	4.5	5.2	5.0	4.7	3.5	3.1	4.5	5.8	6.4	4.7	3.6	2.9	4.2
Precipitation	PrecAvg	94.4	8.9	8.2	9.6	10.4	11.3	7.4	5.6	6.5	4.6	5.6	9.1	7.6
	PrecMax	126.3	16.1	12.9	22.2	19.4	23.1	22.9	18.7	12.7	14.6	14.2	13.2	11.0
	PrecMin	58.0	3.0	4.1	3.9	4.2	2.7	1.6	0.8	0.5	0.2	1.0	3.0	3.7
	PrecStd	21.8	3.7	3.0	4.5	4.3	4.7	5.3	4.8	3.5	3.8	4.0	3.0	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.3	92.4	92.1	90.7	89.2	87.8	85.3	85.0	87.6	89.2	91.4	92.3
		MCWB	80.9	80.8	80.8	80.4	80.3	80.2	78.3	78.1	78.5	79.2	80.3	80.9
	2%	DB	90.6	90.7	90.5	89.4	88.1	86.3	84.2	83.6	86.1	88.0	89.6	90.8
		MCWB	80.1	80.1	80.3	80.1	79.9	79.3	77.9	77.4	78.5	79.0	79.8	80.4
	5%	DB	89.2	89.3	89.1	88.2	87.3	85.4	83.2	82.8	84.6	86.9	88.5	89.5
		MCWB	79.6	79.7	79.9	79.7	79.7	78.9	77.4	77.1	78.0	78.8	79.5	80.0
	10%	DB	88.0	88.0	87.7	87.2	86.4	84.3	82.3	81.9	83.4	85.9	87.5	88.3
		MCWB	79.3	79.1	79.2	79.4	79.4	78.5	76.9	76.7	77.5	78.6	79.2	79.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	81.7	81.9	81.7	81.3	81.0	79.1	78.9	79.8	80.6	81.7	81.9
		MCDB	90.1	90.2	89.4	88.4	87.4	85.8	83.4	83.4	85.2	87.1	90.1	89.6
	2%	WB	80.9	80.8	80.9	80.9	80.7	79.9	78.3	78.1	79.0	79.9	80.8	81.0
		MCDB	88.4	89.0	88.4	87.5	86.6	85.0	82.7	82.4	84.4	86.0	87.7	88.6
	5%	WB	80.2	80.1	80.3	80.3	80.1	79.1	77.8	77.4	78.3	79.4	80.1	80.6
		MCDB	87.3	87.7	87.4	86.8	85.7	84.2	82.1	81.6	83.4	85.3	86.5	88.0
	10%	WB	79.7	79.5	79.7	79.8	79.8	78.7	77.3	77.0	77.7	78.9	79.7	80.0
		MCDB	86.5	86.6	86.4	85.8	85.1	83.5	81.3	81.1	82.3	84.6	85.9	86.9
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.7	11.3	11.0	9.7	7.5	6.2	6.0	7.3	9.7	11.5	11.5
		MCDBR	12.6	12.8	12.7	12.1	11.0	8.9	7.3	7.0	9.5	11.6	12.8	12.8
		MCWBR	5.2	5.1	5.0	5.0	4.6	4.1	3.7	3.6	4.2	4.9	5.1	5.1
	5% WB	MCDBR	11.7	12.4	11.9	11.1	10.1	8.4	7.0	6.7	8.8	10.5	11.9	12.0
MCWBR		5.2	5.2	5.0	4.9	4.5	4.2	3.7	3.6	4.2	4.6	5.1	5.0	
Clear-Sky Solar Irradiance	taub		0.400	0.399	0.396	0.397	0.409	0.419	0.419	0.426	0.434	0.457	0.440	0.414
	taud		2.533	2.528	2.538	2.543	2.510	2.499	2.495	2.463	2.438	2.343	2.401	2.484
	Ebn at Noon		297	297	294	286	275	267	269	274	279	277	282	291
	Edh at Noon		35	35	35	33	33	33	33	36	37	42	39	36
All-Sky Solar Radiation	RadAvg		1622	1663	1635	1595	1533	1372	1322	1407	1539	1679	1682	1607
	RadStd		86	78	56	73	78	90	80	81	88	90	69	77

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page